Problem 7.20

At an effective annual interest rate of i, $i > 0$, each of the following two sets of payments has present value K:

(i) A payment of 121 immediately and another payment of 121 at the end of one year.

(ii) A payment of 144 at the end of two years and another payment of 144 at the end of three years.

Calculate K.

Solution

Let

$$v = \frac{1}{1+i}$$

be the annual discount factor.

Then the present value of set (i) is

$$K = 121 + 121v.$$

The present value of set (ii) is

$$K = 144v^2 + 144v^3.$$

Since both present values are equal,

$$121 + 121v = 144v^2 + 144v^3.$$

Factor both sides:

$$121(1+v) = 144v^2(1+v).$$

Because $v > 0$, we have $1 + v > 0$, so we may cancel $1 + v$ to obtain

$$121 = 144v^2.$$

Thus,

$$v^2 = \frac{121}{144}, \qquad v = \frac{11}{12},$$

since $v > 0$.

Now substitute into $K = 121 + 121v$:

$$K = 121 + 121\left(\frac{11}{12}\right) = 121\left(\frac{23}{12}\right) = \frac{2783}{12}.$$

Hence,

$$K \approx 231.9167.$$

Therefore,

$$\boxed{K \approx 231.92}.$$